Ageing

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Multimorbidity defines glycaemic individuality in ageing

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A recent study published in *Nature Metabolism* finds that the co-occurrence of chronic diseases, rather than any single condition, drives the progressive deterioration of glycaemic control in older adults, reframing glucose management as a multimorbidity-informed challenge.

The questions of why older adults with similar fasting glucose respond differently to the same meal[1,2] and why nocturnal hypoglycaemia is more common among some older adults with diabetes than others[3,4] have long intrigued clinicians and nutrition scientists. In this issue of *Nature Metabolism*, Zhang and colleagues[5] provide compelling evidence that the cumulative burden of multiple chronic diseases, rather than diabetes status alone, is associated with personalized glycaemic trajectories.

The Guangzhou Nutrition and Health Study (GNHS) followed 1,398 participants for a median of 12 years, tracking 21 diseases across seven organ systems. At the latest follow-up (mean age 69 years), each individual underwent continuous glucose monitoring and standardized meal challenges, generating more than 16,000 daily glycaemic measurements. This longitudinal design combined disease trajectories, real-time glucose dynamics and repeated serum proteomics, and enabled analysis of how multimorbidity accumulation relates to glycaemic variation between individuals.

The study found a dose-dependent association between a systemic multimorbidity index (MMI system) and increased glucose variability, higher hyperglycaemia risk and stronger individualized postprandial glucose responses. Crucially, the cumulative burden of multiple diseases, rather than any single condition, appears to be the key driver of glucose instability in older adults. This consistent trend, even observed in populations without diabetes, highlights the effect of multimorbidity on metabolic health.

Specifically, seven interconnected disorders formed a tightly clustered metabolic multimorbidity network, including type 2 diabetes, obesity, metabolic dysfunction-associated steatotic liver disease (MASLD), hypertension, gout, dyslipidaemia and chronic kidney disease[6,7]. Notably, dyslipidaemia was identified as the earliest occurring condition and a potential trigger for the cascade of metabolic disturbances. This finding underscores the critical importance of early lipid management to prevent the development of multimorbidity and glucose dysregulation.

Equally importantly, the relationship extends to personalized responses to identical meals. Using a previously validated personalized glycaemic sensitivity (PGS) index[8], the authors show that multimorbidity severity predicts heightened postprandial excursions in a dose-dependent manner. Healthy older adults resembled young control individuals in their meal responses, whereas those with multimorbidity showed elevated responses. Duration of MASLD, hypertension and adiposity correlated with higher PGS, whereas chronic kidney disease progression unexpectedly associated with reduced postprandial spikes — and elevated hypoglycaemia risk, particularly at night. This observation points to a potential bidirectional relationship between glucose and disease. Chronic kidney disease may blunt counter-regulatory responses or alter medication clearance, predisposing to hypoglycaemia. Conversely, recurrent hypoglycaemic episodes could affect renal function through repeated sympathetic activation or haemodynamic stress. Future studies using longitudinal intervention designs could help establish whether glucose and multimorbidity engage in reciprocal interactions. For example, testing whether dietary adjustments guided by continuous glucose monitoring can modify disease trajectories, or whether prevention of hypoglycaemia slows kidney function decline.

Ageing provides essential context for these findings. Chronological age correlated modestly with multimorbidity accumulation, but the key factor was the divergence of disease trajectories with advancing years. Some individuals aged without accumulating many conditions; others followed accelerated multimorbidity paths. This heterogeneity translated to widening gaps in glycaemic control. Ageing thus functions as an amplifier, revealing and exacerbating underlying differences in physiological resilience shaped by lifelong exposures and genetic susceptibility[9,10]. Nevertheless, the precise molecular mechanisms through which ageing modifies the relationship between multimorbidity and glucose homeostasis remain to be fully elucidated. Whether ageing acts primarily as a time-dependent accumulator of disease or actively remodels physiological set-points to alter glucose sensitivity warrants deeper investigation.

The study also investigated the molecular mechanisms linking multimorbidity to glucose sensitivity. Through longitudinal analysis of large-scale serum proteomic data, Zhang and colleagues identified 218 proteins associated with systemic multimorbidity dynamics, of which 131 were validated in an external cohort. These proteins were enriched for lipid homeostasis pathways — lipid transport, cholesterol metabolism and lipoprotein remodelling. Protein co-expression networks progressively fragmented with increasing multimorbidity severity, suggesting loss of coordinated metabolic regulation. A protein score comprising 44 multimorbidity and PGS-associated proteins statistically mediated 12.9% of the association between disease burden and personalized glycaemic sensitivity. Vitronectin showed positive association with PGS, whereas sex hormone-binding globulin, apolipoprotein F and gelsolin were negatively associated, with each representing potential mechanistic nodes connecting multi-organ pathology to glucose regulation (Fig. 1).

This study demonstrates potential for clinical application. A machine learning model integrating multimorbidity information with

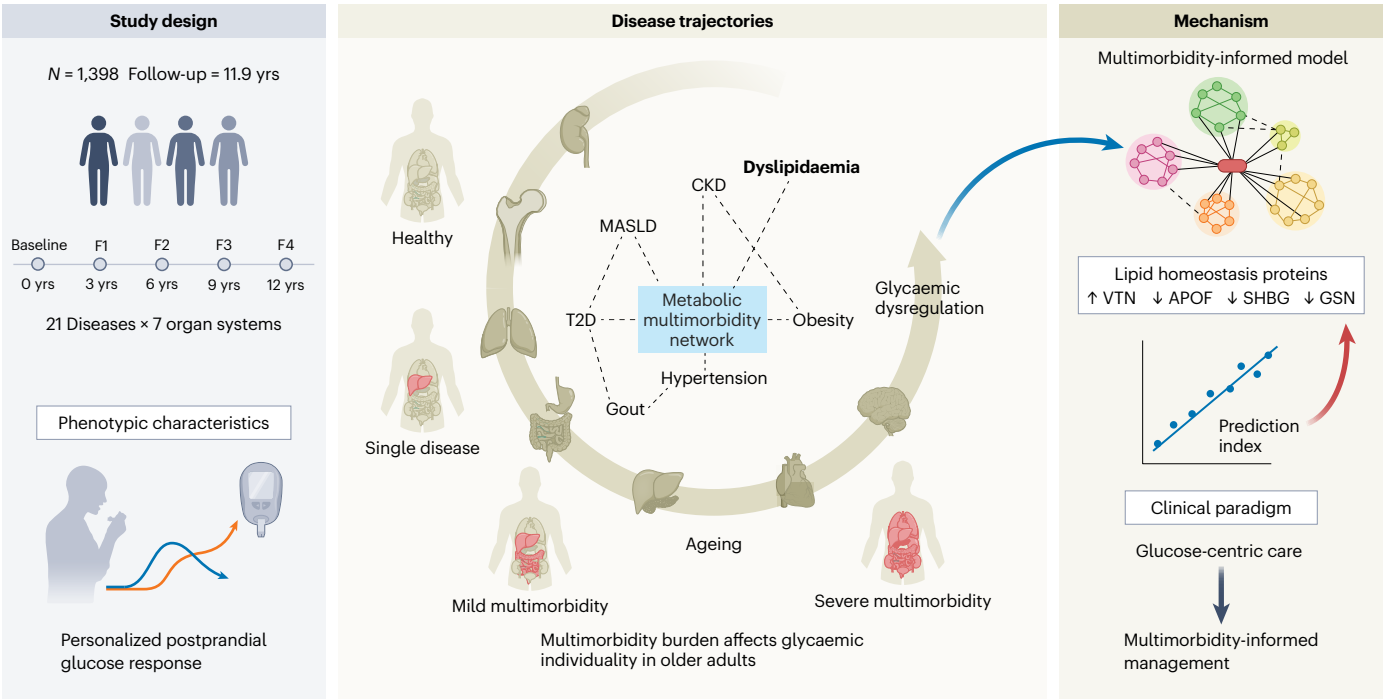

Fig. 1 | Multimorbidity burden drives personalized glycaemic dysregulation in ageing. The accumulation of multiple chronic diseases, particularly those within a metabolic network, correlates with progressive and individualized deterioration of glucose control in older adults. The work by Zhang and colleagues establishes that multimorbidity burden shapes glycaemic trajectories, shifting the clinical paradigm from glucose-centric care to multimorbidity-informed management. APOF, apolipoprotein F; CKD, chronic kidney disease; GSN, gelsolin; SHBG, sex hormone-binding globulin; T2D, type 2 diabetes; VTN, Vitronectin; yrs, years.

traditional glycaemic markers demonstrated advantages in predicting average glucose status, glucose variability and particularly the risk of nocturnal hypoglycaemia. Furthermore, the model showed improvements in predicting individualized postprandial glucose responses and PGS, with average predictive performance enhanced by 35% to 43.8%. This finding provides a powerful tool for clinical practice, offering the potential for more refined glucose management and improved hypoglycaemia risk identification in older adults[11].

Several new directions for future intervention strategies emerge from this work. First, the early emergence of dyslipidaemia in the disease cascade suggests that lipid management in middle-aged adults with isolated dyslipidaemia could be tested for its potential to attenuate later multimorbidity accumulation and glycaemic decline. Second, proteomic signatures could be evaluated as stratification tools in precision nutrition trials. Third, multimorbidity-informed algorithms embedded in continuous glucose monitoring systems might provide real-time hypoglycaemia alerts calibrated to individual disease burden. Fourth, given the centrality of lipid homeostasis proteins, interventions targeting lipid metabolism whether dietary, pharmacological or lifestyle could be investigated for their potential to simultaneously affect multimorbidity progression and glycaemic control.

Several questions remain. The observational design precludes causal certainty, although longitudinal proteomics and mediation analysis strengthen mechanistic inference. Future integration with multi-omic approaches might reveal additional mediators, such as gut microbiota, metabolomics, epigenetics, and so on. Furthermore, external validation in non-Chinese populations is needed to establish generalizability[12]. Crucially, distinguishing whether specific molecular drivers of multimorbidity directly cause glycaemic deterioration or whether these represent parallel consequences of a shared ageing process requires experimental models and mechanistic studies capable of disentangling directionality from correlation.

This work reframes glucose management in older adults by suggesting that glycaemic control might be understood partly as a reflection of systemic multimorbidity burden and represents an important step in understanding the complex interplay between ageing, multimorbidity and metabolic health[13]. For older adults with multiple chronic conditions, stabilizing overall disease trajectory – potentially beginning with lipid optimization – may warrant investigation alongside glucose-lowering approaches. Zhang and colleagues provide both a conceptual framework and molecular tools to pursue this integrated perspective. The path to precision nutrition in ageing may lie not through single diseases but through the complex landscape of multimorbidity itself.

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Competing interests

The authors declare no competing interests.